

Qt Advanced Course

TT00GD57 — Kick-off Session

Monday 16.3.2026, 16:00–17:00

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Agenda

- **Course Overview** Timeline, mentoring model, what to expect
- **Grading** 0–5 scale rubric, baseline requirements
- **Project Scoping** What 5 credits means, the two-sided architecture
- **Qt Installation** Open source vs. Educational license
- **Next Steps** Background survey, Friday session, one-on-ones

Course Overview & Timeline

- **16.3.** Kick-off (this session)
- **20.3.** Project plan preparation (Fri 13:00)
- **Week of 23.3.** Individual one-on-one sessions
- **1.–10.4.** No mentoring sessions available
- **11.4.** Mid-course checkpoint
- **Week 19** Final presentations + feedback

Mentoring Model

- Weekly one-on-one sessions
- Sessions can be quick — no minimum
- Email me anytime for questions

Student Deliverable

- Independent Qt 6 project
- QML UI + C++ backend
- Network protocol
- 10–15 min presentation (week 19)

Grading: Baseline Requirements

Grading is 0–5. These must be met to pass:

- App runs on the chosen target platform
- QML UI connected to a C++ backend
- At least one network protocol used (or justified exception)
- Code is structured and readable
- 10–15 min presentation at wrap-up (week 19)

Your choice of target platform (desktop, emulator, Android, embedded) does not affect grading.

You should be able to explain your implementation decisions and how your code works.

Grading Rubric (0–5)

Dimension	Weight	1 (minimum pass)	3 (solid)	5 (excellent)
Functionality	30%	End-to-end hello world — app runs, one feature works	Multiple features working reliably	Polished, handles edge cases gracefully
Architecture	25%	QML + C++ connected, readable code	Clean separation, proper signals/slots	Well-designed, reusable components
Networking	20%	One protocol at hello-world level	Meaningful use, data flows both ways	Error handling, reconnection logic
UI / UX	15%	Basic QML displays data	Responsive, clear navigation	Polished, animations, adaptive
Presentation	10%	Demo shown, project explained	Clear structure, good demo	Engaging, insightful reflection

Grade 0 = requirements not met or project not submitted

Project Scoping: What 5 Credits Means

- **~135 hours** total including learning time
- **60–80 hours** actual coding and building
- **The rest** toolchain setup, learning protocols, debugging deployment

Start small, extend later

- MVP first: one screen, one data flow, one protocol working end-to-end
- Then add: more screens, error handling, polish
- A working simple app beats a broken ambitious one

The Two-Sided Architecture

Every project has two communicating sides:

Your Qt Quick App

- QML UI
- C++ backend
- Runs on your target device / emulator



protocol

The "Other Side"

- MQTT broker (Mosquitto)
- Python script
- Public test server / API
- Another Qt app

Think about both sides when defining your project!

Common Scoping Mistakes

- Trying to use too many protocols at once
- Over-designing the UI before backend works
- Underestimating toolchain setup time
- Skipping the "other side" (no server, no data)

Example: MQTT Temperature Dashboard

- **Qt Quick app** shows live temperature readings
- **Mosquitto broker** runs locally or in the cloud
- **Python script** publishes fake sensor data

QML UI + C++ backend + network protocol. Extend from there.

Qt Installation

Install at least Qt 6.10.2 with the Desktop target

Option 1: Open Source

- Free, no sign-up required
- Sufficient for desktop + Android targets
- qt.io/development/download-qt-installer-oss

Option 2: Educational License

- Required for: Boot to Qt (embedded Linux)
- Required for: Qt for MCUs
- Sign up with your metropolia.fi email
- Select "Qt Edu for Developers"
- qt.io/development/qt-educational-license

Note: If you have open source Qt installed, uninstall before installing educational edition.

Action: Have Qt 6.10.2 and Qt Creator installed and working before Friday's session!

Next Steps

- **Background survey** Fill in by end of Thursday 19.3.
- **Friday 20.3. at 13:00** Project plan preparation session
- **Week of 23.3.** Individual one-on-one sessions for project plans
- **Install Qt** Have Qt 6.10.2 + Qt Creator ready by Friday

Start thinking about your project idea — we'll refine it together!